EE 533 Information Theory

Week 07

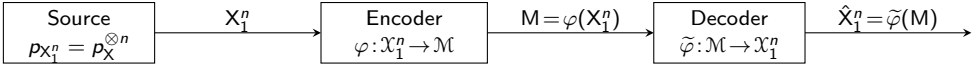
Barış Nakiboğlu

Department of Electrical and Electronics Engineering
Middle East Technical University

April 07, 2025

Channel Coding Problem

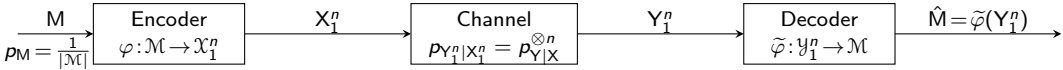
A length n source code $(\varphi, \widetilde{\varphi})$ for an i.i.d. source $\mathsf{X}_1, \ldots, \mathsf{X}_n$:



$$R = \frac{1}{n} \log |\mathcal{M}|, \qquad P_{\mathsf{e}}^{(n)} = \mathbf{P}\left(\hat{\mathsf{X}}_1^n \neq \mathsf{X}_1^n\right).$$

Source Coding Problem: What is the minimum value of R that can ensure $P_{\mathsf{e}}^{(n)} \to 0$ as $n \to \infty$ for the optimal choice of the code $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$ for each n.

A length n channel code $(\varphi, \widetilde{\varphi})$ for an memoryless/product channel:



$$R = \frac{1}{n} \log |\mathcal{M}|, \qquad P_{\mathsf{e}}^{(n)} = \mathbf{P}\left(\hat{\mathsf{M}} \neq \mathsf{M}\right).$$

Channel Coding Problem: What is the maximum value of R that can ensure $P_{\mathsf{e}}^{(n)} \to 0$ as $n \to \infty$ for the optimal choice of the code $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$ for each n.

DMCs: Stationary Discrete Product Channels

Definition (Discrete Channel)

A discrete channel consists of a countable input alphabet $\mathcal{X}$, a countable output alphabet $\mathcal{Y}$, and a conditional pmf $p_{\mathsf{Y}|\mathsf{X}}$ determining distribution of the channel output Y given the channel input X.

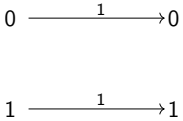
- Each conditional pmf $p_{\mathsf{Y}|\mathsf{X}}$ can be expressed equivalently by a function $W : \mathcal{X} \to \mathcal{P}(\mathcal{Y})$, satisfying $p_{\mathsf{Y}|\mathsf{X}}(y|x) = W(y|x)$ for all $x \in \mathcal{X}, y \in \mathcal{Y}$, where $\mathcal{P}(\mathcal{Y})$ is the set of all pmf's on $\mathcal{Y}$,
- If $\mathcal{X}$ and $\mathcal{Y}$ are finite sets W can be expressed as a matrix (probability transition matrix)

$$W = \begin{bmatrix} W(x_1) \\ \vdots \\ W(x_k) \end{bmatrix} = \begin{bmatrix} W(y_1|x_1) & \cdots & W(y_j|x_1) \\ \vdots & & \vdots \\ W(y_1|x_k) & \cdots & W(y_j|x_k) \end{bmatrix}$$

- A discrete channel $p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}$ is *memoryless* iff it satisfies $p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}(y_1^n|x_1^n) = \prod_{t=1}^n p_{\mathsf{Y}_t|\mathsf{X}_t}(y_t|x_t)$
  - Such a channel is stationary if $p_{\mathsf{Y}_t|\mathsf{X}_t} = p_{\mathsf{Y}_1|\mathsf{X}_1}$ for all $t \in \{1, \ldots, n\}$.
  - Note that according to the definition of memorylessness given above, which is standard, the input set of the channel $p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}$ can be a proper subset of $\mathcal{X}_1^n$.
  - If the input set of a memoryless channel $p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}$ is $\mathcal{X}_1^n$, then it is called a product channel.
  - Discrete stationary product channels are commonly called DMCs (discrete memoryless channels), with a clear abuse of jargon.

Discrete Channel Examples - I

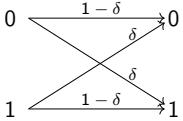
Example (Noiseless Binary Channel)



$$\mathcal{X} = \{0, 1\}$$
$$\mathcal{Y} = \{0, 1\}$$

$$W = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

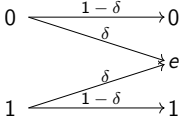
Example (Binary Symmetric Channel)



$$\mathcal{X} = \{0, 1\}$$
$$\mathcal{Y} = \{0, 1\}$$

$$W = \begin{bmatrix} 1-\delta & \delta \\ \delta & 1-\delta \end{bmatrix}$$

Example (Binary Erasure Channel)

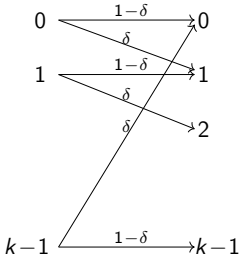


$$\mathcal{X} = \{0, 1\}$$
$$\mathcal{Y} = \{0, e, 1\}$$

$$W = \begin{bmatrix} 1-\delta & \delta & 0 \\ 0 & \delta & 1-\delta \end{bmatrix}$$

Discrete Channel Examples - II

Example (Noisy Typewriter)



$$\mathcal{X} = \{0, 1, \ldots, k-1\}$$
$$\mathcal{Y} = \{0, 1, \ldots, k-1\}$$
$$p_{Y|X} = \begin{cases} 1-\delta & y = x \\ \delta & y = x+1 \bmod k \end{cases}$$

$$W = \begin{bmatrix} 1-\delta & \delta & 0 & \cdots & 0 \\ 0 & 1-\delta & \delta & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \delta & 0 & 0 & \cdots & 1-\delta \end{bmatrix}$$

Example (Additive Geometric Noise Channel)

$$Y = X + Z$$
$$p_Z = \begin{cases} (1-\delta)\delta^z & z \in \mathbb{Z}_{\geq 0} \\ 0 & \text{else} \end{cases}$$

$$\mathcal{X} = \mathbb{Z}_{\geq 0}$$
$$\mathcal{Y} = \mathbb{Z}_{\geq 0}$$
$$p_{Y|X} = p_Z(y-x)$$

$$W = \begin{bmatrix} 1-\delta & (1-\delta)\delta & (1-\delta)\delta^2 & \cdots \\ 0 & 1-\delta & (1-\delta)\delta & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Shannon Capacity and Shannon Radius I

Definition (Shannon Capacity)

For any discrete channel $p_{Y|X}$, the Shannon Capacity $C\left(p_{Y|X}\right)$ is defined as

$$C\left(p_{Y|X}\right) := \sup_{p_X} I(X;Y)\,,$$

Using the the variational characterization of mutual information

$$\begin{aligned}
C\left(p_{Y|X}\right) &= \sup_{p_X} \inf_{p_Z} D\left(p_X p_{Y|X} \middle\| p_X p_Z\right) && \text{by } I(X;Y) = \inf_{p_Z} D\left(p_X p_{Y|X} \middle\| p_X p_Z\right) \\
&\leq \inf_{p_Z} \sup_{p_X} D\left(p_X p_{Y|X} \middle\| p_X p_Z\right) && \text{by the max-min inequality (wikipedia)} \\
&\leq \inf_{p_Z} \sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Z\right) && \text{by } D\left(p_X p_{Y|X} \middle\| p_X p_Z\right) \leq \sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Z\right) \\
&= \inf_{p_Z} S\left(p_{Y|X}, p_Z\right) && \text{by the definition of relative Shannon radius} \\
&= S\left(p_{Y|X}\right) && \text{by the definition of Shannon radius}
\end{aligned}$$

H.W. Prove that $\sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Z\right) = \sup_{p_X} D\left(p_X p_{Y|X} \middle\| p_X p_Z\right)$. This identity implies that the second inequality in above chain is an equality.

Shannon Capacity and Shannon Radius

Lemma

For any discrete channel $p_{Y|X}$ with a finite input set, $\exists p_{X^\star}$ s.t. $I(X^\star;Y) = C\left(p_{Y|X}\right)$. Furthermore, $\exists! p_{Y^\star}$ satisfying $p_{Y^\star}(y) = \sum_x p_X(x)\, p_{Y|X}(y|x)$ for any p_X satisfying $I(X;Y) = C\left(p_{Y|X}\right)$ and

$$D\left(p_{Y|X}(\cdot|x) \middle\| p_{Y^\star}\right) \leq C\left(p_{Y|X}\right) \qquad \forall x \in \mathcal{X} \qquad (*)$$

- $S\left(p_{Y|X}\right) \leq C\left(p_{Y|X}\right)$ because $S\left(p_{Y|X}, p_{Y^\star}\right) \leq C\left(p_{Y|X}\right)$ by (*) and $S\left(p_{Y|X}\right) \leq S\left(p_{Y|X}, p_Y\right)\ \forall p_Y$. Thus for finite $\mathcal{X}$ case $C\left(p_{Y|X}\right) = S\left(p_{Y|X}\right)$ by $S\left(p_{Y|X}\right) \geq C\left(p_{Y|X}\right)$, which holds for any $\mathcal{X}$.
- We can confirm $I(X;Y) = C\left(p_{Y|X}\right)$ for a given p_X, and confirm the value of $C\left(p_{Y|X}\right)$ using the lemma. This is interesting because $C\left(p_{Y|X}\right)$ does not have a closed form expression.
  - We can calculate $I(X;Y)$ and p_Y for p_X because they have closed form expressions.
  - $\sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Y\right) \leq I(X;Y)$ for a p_X iff $C\left(p_{Y|X}\right) = I(X;Y)$, because
    If $I(X;Y) = C\left(p_{Y|X}\right)$ then $\sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Y\right) \leq I(X;Y)$ by the lemma.
    If $I(X;Y) < C\left(p_{Y|X}\right)$ then $\sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Y\right) > I(X;Y)$ because

$$\begin{aligned} \sup_{x\in\mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Y\right) &\geq S\left(p_{Y|X}\right) && \text{by the definition of } S\left(p_{Y|X}\right) \\ S\left(p_{Y|X}\right) &= C\left(p_{Y|X}\right) && \text{by the lemma.} \end{aligned}$$

Proof.

If $\mathcal{X}$ is a finite set then $I(\mathrm{X};\mathrm{Y})$ is continuous in p_{X} on the probability simplex, which is a closed and bounded (i.e. compact) set, in $\mathbb{R}^{|\mathcal{X}|}$. Thus $p_{\mathrm{X}^\star}$ exists by *the extreme value theorem*.

$$\begin{aligned} I(\mathrm{X}_\lambda;\mathrm{Y}) &= D\big(p_{\mathrm{X}_\lambda} p_{\mathrm{Y}|\mathrm{X}} \big\| p_{\mathrm{X}_\lambda} p_{\mathrm{Y}_\lambda}\big) && \text{for } p_{\mathrm{X}_\lambda} = (1-\lambda)p_{\mathrm{X}^\star} + \lambda p_{\mathrm{X}} \\ &= (1-\lambda) D\big(p_{\mathrm{X}^\star} p_{\mathrm{Y}|\mathrm{X}} \big\| p_{\mathrm{X}^\star} p_{\mathrm{Y}_\lambda}\big) + \lambda D\big(p_{\mathrm{X}} p_{\mathrm{Y}|\mathrm{X}} \big\| p_{\mathrm{X}} p_{\mathrm{Y}_\lambda}\big) \\ &= (1-\lambda)(I(\mathrm{X}^\star;\mathrm{Y}) + D(p_{\mathrm{Y}^\star}\|p_{\mathrm{Y}_\lambda})) + \lambda D\big(p_{\mathrm{X}} p_{\mathrm{Y}|\mathrm{X}} \big\| p_{\mathrm{X}} p_{\mathrm{Y}_\lambda}\big) \\ &\geq (1-\lambda) C\left(p_{\mathrm{Y}|\mathrm{X}}\right) + \lambda(I(\mathrm{X};\mathrm{Y}) + D(p_{\mathrm{Y}}\|p_{\mathrm{Y}_\lambda})) \end{aligned}$$

Then $I(\mathrm{X}_\lambda;\mathrm{Y}) \leq C\left(p_{\mathrm{Y}|\mathrm{X}}\right)$ imply

$$C\left(p_{\mathrm{Y}|\mathrm{X}}\right) \geq I(\mathrm{X};\mathrm{Y}) + D(p_{\mathrm{Y}}\|p_{\mathrm{Y}_\lambda})$$

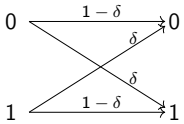
$\liminf_{\lambda\downarrow 0} D(p_{\mathrm{Y}}\|p_{\mathrm{Y}_\lambda}) \geq D(p_{\mathrm{Y}}\|p_{\mathrm{Y}^\star})$ by the lower semi-continuity of KLD because $\lim_{\lambda\downarrow 0} p_{\mathrm{Y}_\lambda} = p_{\mathrm{Y}^\star}$

$$C\left(p_{\mathrm{Y}|\mathrm{X}}\right) \geq I(\mathrm{X};\mathrm{Y}) + D(p_{\mathrm{Y}}\|p_{\mathrm{Y}^\star})$$

Then $I(\mathrm{X};\mathrm{Y}) = C\left(p_{\mathrm{Y}|\mathrm{X}}\right)$ implies $D(p_{\mathrm{Y}}\|p_{\mathrm{Y}^\star}) = 0$ and hence $p_{\mathrm{Y}} = p_{\mathrm{Y}^\star}$
Furthermore, $D\big(p_{\mathrm{Y}|\mathrm{X}}(\cdot|x)\big\|p_{\mathrm{Y}^\star}\big) \leq C\left(p_{\mathrm{Y}|\mathrm{X}}\right)$ for all $x \in \mathcal{X}$ because $p_{\mathrm{Y}}(y) = p_{\mathrm{Y}|\mathrm{X}}(y|x)$ for $p_{\mathrm{X}}(z) = \mathbb{1}_{\{z=x\}}$ for all $x \in \mathcal{X}$ $\square$

Shannon Capacities of Certain Discrete Channels

Example (Binary Symmetric Channel)

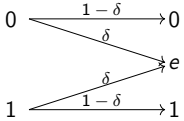


$$W = \begin{bmatrix} 1-\delta & \delta \\ \delta & 1-\delta \end{bmatrix}$$

$$C(W) = \log 2 - \hbar(\delta)$$

$$p_{Y^\star} = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}$$

Example (Binary Erasure Channel)



$$W = \begin{bmatrix} 1-\delta & \delta & 0 \\ 0 & \delta & 1-\delta \end{bmatrix}$$

$$C(W) = (1-\delta)\log 2$$

$$p_{Y^\star} = \begin{bmatrix} \frac{1-\delta}{2} & \delta & \frac{1-\delta}{2} \end{bmatrix}$$

Example (Noisy Typewriter)

$$p_{Y|X} = \begin{cases} 1-\delta & y = x \\ \delta & y = x+1 \bmod k \end{cases}$$

$$W = \begin{bmatrix} 1-\delta & \delta & \cdots & 0 \\ 0 & 1-\delta & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \delta & 0 & \cdots & 1-\delta \end{bmatrix}$$

$$C(W) = \log k - \hbar(\delta)$$

$$p_{Y^\star} = \begin{bmatrix} \frac{1}{k} & \frac{1}{k} & \cdots & \frac{1}{k} \end{bmatrix}$$

Symmetric Channels

- $p_{Y|X}$ is *symmetric* iff all rows of the probability transition matrix, i.e., $p_{Y|X}(\cdot|x)$'s, are permutations of each other and all columns, i.e. $p_{Y|X}(y|\cdot)$'s, are permutations of each other.
  Example: BSC, noisy typewriter channel.
- $p_{Y|X}$ is *weakly symmetric* iff all rows of the probability transition matrix, i.e., $p_{Y|X}(\cdot|x)$'s, are permutations of each other and all column sums, i.e., $\sum_x p_{Y|X}(y|x)$'s, are equal.
  Example: All symmetric channels.

  There are weakly symmetric channels that are not symmetric, e.g. $\begin{bmatrix} 2/3 & 1/3 & 0 \\ 0 & 1/3 & 2/3 \end{bmatrix}$.
- $p_{Y|X}$ is *Gallager-symmetric* if the output set $\mathcal{Y}$ can be partitioned into disjoint subsets $\mathcal{Y}_1, \ldots, \mathcal{Y}_k$ in such a way that within each $|\mathcal{X}| \times |\mathcal{Y}_k|$ transition matrix all rows are permutations of each other and all columns are permutations of each other.
  Example: All symmetric channels, BEC with an arbitrary δ.
- Not all Gallager-symmetric channels are weakly symmetric, e.g. $\begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}$.
- H.W. Construct a weakly-symmetric channel that is not Gallager-symmetric rows of whose transition probability matrix are permutations of $\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} & 0 \end{bmatrix}$.

Shannon Capacity of Symmetric Channels

$p_{Y|X}$ is *weakly Gallager-symmetric* if the output set $\mathcal{Y}$ can be partitioned into disjoint subsets $\mathcal{Y}_1, \ldots, \mathcal{Y}_k$ in such a way that within each $|\mathcal{X}| \times |\mathcal{Y}_k|$ transition matrix all rows are permutations of each other and all columns sums, i.e. $\sum_x p_{Y|X}(y|x)$, are equal.

Example: All weakly symmetric channels, all Gallager-symmetric channels.

Lemma

For any weakly Gallager-symmetric discrete channel $p_{Y|X}$ $C\left(p_{Y|X}\right) = I(X^\star; Y)$ for $p_{X^\star}(x) = 1/|\mathcal{X}|$, i.e. for an $X^\star$ that is uniformly distributed on the input set $\mathcal{X}$. Furthermore, if $p_{Y|X}$ is also weakly symmetric then $C\left(p_{Y|X}\right) = \ln|\mathcal{Y}| - H(Y|X = x)$.

Proof.

As a result of weak Gallager-symmetry $p_{Y^\star}$ will be constant within each $\mathcal{Y}_k$. Thus

$$p_{Y^\star}(y) = \sum_{t=1}^{k} \frac{\beta_t}{|\mathcal{Y}_t|} \mathbb{1}_{\{y \in \mathcal{Y}_t\}} \qquad \text{for } \beta_1, \ldots, \beta_k \text{ s.t. } \sum_{t=1}^{k} \beta_t = 1$$

$$D\left(p_{Y|X}(\cdot|x) \middle\| p_{Y^\star}\right) = \sum_{t=1}^{k} \beta_t \ln \frac{|\mathcal{Y}_t|}{\beta_t} - H(Y|X = x) \qquad \forall x \in \mathcal{X}$$

$$D\left(p_{Y|X}(\cdot|x) \middle\| p_{Y^\star}\right) = I(X^\star; Y) \qquad \forall x \in \mathcal{X}$$

For weakly symmetric channels $k = 1$, hence $\mathcal{Y}_1 = \mathcal{Y}$ and $\beta_1 = 1$.

Equality of Shannon Capacity and Radius For Arbitrary Discrete Channels

Lemma ([Kem74])

For any discrete channel $p_{Y|X}$

$$\begin{aligned}\sup_{p_X} \inf_{p_Z} D\left(p_X p_{Y|X} \middle\| p_X p_Z\right) &= \inf_{p_Z} \sup_{p_X} D\left(p_X p_{Y|X} \middle\| p_X p_Z\right) \\ &= \inf_{p_Z} \sup_{x \in \mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_Z\right)\end{aligned}$$

If $C\left(p_{Y|X}\right) < \infty$*, then there exists a unique* $p_{Y^\star} \in \mathcal{P}(\mathcal{Y})$*, called the Shannon Center, satisfying*

$$\begin{aligned}C\left(p_{Y|X}\right) &= \sup_{p_X} D\left(p_X p_{Y|X} \middle\| p_X p_{Y^\star}\right) \\ &= \sup_{x \in \mathcal{X}} D\left(p_{Y|X}(\cdot|x) \middle\| p_{Y^\star}\right)\end{aligned}$$

Furthermore, for any sequence of input distributions $\{p_{X_i}\}_{i \in \mathbb{Z}_+}$ *satisfying* $\lim_{i \to \infty} I(X_i; Y) = C\left(p_{Y|X}\right)$ *corresponding output distributions satisfy* $\lim_{i \to \infty} \|p_{Y_i} - p_{Y^\star}\|_1 = 0$.

Proof idea due to Kemperman.

- Apply lemma for the finite $\mathcal{X}$ case via a sequence of input distributions $\{p_{X_i}\}_{i \in \mathbb{Z}_+}$ satisfying $\sum_x \mathbb{1}_{\{p_{X_i}(x) > 0\}} < \infty$ for all i and $\lim_{i \to \infty} I(X_i; Y) = C\left(p_{Y|X}\right)$
- Invoke the lower semicontinuity of KLD.

Shannon Capacity and Center of Product Channels

Lemma *For any discrete product channel* $p_{Y_1^n|X_1^n}$

$$C\left(p_{Y_1^n|X_1^n}\right) = \sum_{t=1}^{n} C\left(p_{Y_t|X_t}\right) \tag{*}$$

$$p_{Y_1^{n\star}}(y_1^n) = \prod_{t=1}^{n} p_{Y_t^\star}(y_t) \tag{**}$$

Proof.

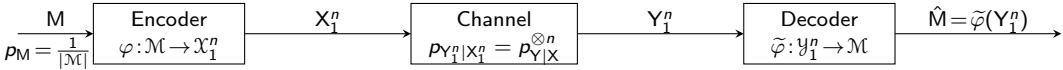
$$\begin{aligned} I(X_1^n; Y_1^n) &= H(Y_1^n) - H\left(Y_1^n|X_1^n\right) \\ &= H(Y_1^n) - \sum_{t=1}^{n} H\left(Y_t|X_t\right) \\ &\leq \sum_{t=1}^{n} \left[H(Y_t) - H\left(Y_t|X_t\right)\right] \quad \text{equality holds iff } Y_1, \ldots, Y_n \text{ are joint independent.} \\ &= \sum_{t=1}^{n} I(X_t; Y_t) \end{aligned}$$

Then (*) holds because, if $X_1, \ldots, X_n$ are joint independent, then so are $Y_1, \ldots, Y_n$.

$$\begin{aligned} D\left(p_{Y_1^n|X_1^n}(\cdot|x_1^n)\middle\| p_{Y_1^\star} \cdots p_{Y_n^\star}\right) &= \sum_{t=1}^{n} D\left(p_{Y_t|X_t}(\cdot|x_t)\middle\| p_{Y_t^\star}\right) \\ &\leq \sum_{t=1}^{n} C\left(p_{Y_t|X_t}\right) \end{aligned}$$

Then (**) holds as a result of (*).

Channel Coding Problem



DMC $p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}(y_1^n|x_1^n) = \prod_{t=1}^n p_{\mathsf{Y}|\mathsf{X}}(y_t|x_t) \qquad \forall x_1^n, y_1^n$

Encoding function $\varphi : \mathcal{M} \to \mathcal{X}_1^n$

Decoding function $\widetilde{\varphi} : \mathcal{Y}_1^n \to \mathcal{M}.$

$$P_{\mathbf{e}}^{(n)} := \mathbf{P}\left(\hat{\mathsf{M}} \neq \mathsf{M}\right) = \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^n) \neq \mathsf{M}),$$

$$P_{\mathbf{e}}^{m} := \mathbf{P}\left(\hat{\mathsf{M}} \neq \mathsf{M} \middle| \mathsf{M} = m\right) = \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^n) \neq m | \mathsf{X}_1^n = \varphi(m)),$$

$$P_{\mathbf{e}}^{(n),max} := \max_{m \in \mathcal{M}} P_{\mathbf{e}}^{m}.$$

- For any $|\mathcal{M}| \in \mathbb{Z}_+$, one can have $P_{\mathbf{e}}^{(n)} \downarrow 0$ as $n \to \infty$. Why?
- How about when $|\mathcal{M}| = \lceil 2^{nR} \rceil$ for some $R > 0$?

Channel Coding Theorem

Theorem

For any DMC $p_{Y|X}$, all rates below the Shannon capacity of $p_{Y|X}$ are achievable: for every rate $R < C\left(p_{Y|X}\right)$, there exists a sequence of $(\lceil 2^{nR} \rceil, n)$ codes satisfying $\limsup_{n\to\infty} P_{\mathbf{e}}^{(n),max} = 0$. Conversely, for any sequence of $(\lceil 2^{nR} \rceil, n)$ codes on $p_{Y|X}$ with a $R > C\left(p_{Y|X}\right)$, the maximum probability of error is bounded away from zero, i.e. $\liminf_{n\to\infty} P_{\mathbf{e}}^{(n),max} > 0$.

Proof Idea for the achievability of any rate R satisfying $R < C\left(p_{Y|X}\right)$.

- Use random coding: choose each entry of each codeword independently with p_X,
- Use typicality decoding: decode to a message iff its codeword is the only codeword in the code that is jointly typical with the channel output. □

Proof Idea for the non-achievability of any rate R satisfying $R > C\left(p_{Y|X}\right)$.

- Fano's inequality,
- Can be strengthened: $\liminf_{n\to\infty} P_{\mathbf{e}}^{(n),av} = 1$. Strong Converse □

A Review of Entropy Typicality I

Definition (Joint Entropy Typicality)

For any pmf $p_{\mathsf{X},\mathsf{Y}}$ and $\gamma > 0$ the (γ-entropy) typical sets $\mathcal{A}_\gamma^{(n)}(\mathsf{X})$, $\mathcal{A}_\gamma^{(n)}(\mathsf{Y})$, and $\mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y})$ are defined as

$$\begin{aligned}
\mathcal{A}_\gamma^{(n)}(\mathsf{X}) &:= \{x_1^n : \text{satisfying (1)}\}, \\
\mathcal{A}_\gamma^{(n)}(\mathsf{Y}) &:= \{y_1^n : \text{satisfying (2)}\}, \\
\mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y}) &:= \{(x_1^n, y_1^n) : \text{satisfying (1), (2), and (3)}\},
\end{aligned}$$

where

$$\left| \frac{1}{n} \log \frac{1}{p_{\mathsf{X}_1^n}(x_1^n)} - H(\mathsf{X}) \right| < \gamma, \tag{1}$$

$$\left| \frac{1}{n} \log \frac{1}{p_{\mathsf{Y}_1^n}(y_1^n)} - H(\mathsf{Y}) \right| < \gamma, \tag{2}$$

$$\left| \frac{1}{n} \log \frac{1}{p_{\mathsf{X}_1^n,\mathsf{Y}_1^n}(x_1^n, y_1^n)} - H(\mathsf{X},\mathsf{Y}) \right| < \gamma. \tag{3}$$

A Review of Entropy Typicality II

$$(x_1^n, y_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathrm{X},\mathrm{Y}) \text{ iff } \begin{cases} 2^{-n(H(\mathrm{X})+\gamma)} \leq \quad p_{\mathrm{X}_1^n}(x_1^n) \quad \leq 2^{-n(H(\mathrm{X})-\gamma)}, \text{and} \\ 2^{-n(H(\mathrm{Y})+\gamma)} \leq \quad p_{\mathrm{Y}_1^n}(y_1^n) \quad \leq 2^{-n(H(\mathrm{Y})-\gamma)}, \text{and} \\ 2^{-n(H(\mathrm{X},\mathrm{Y})+\gamma)} \leq p_{\mathrm{X}_1^n,\mathrm{Y}_1^n}(x_1^n, y_1^n) \leq 2^{-n(H(\mathrm{X},\mathrm{Y})-\gamma)}. \end{cases} \qquad (*)$$

Theorem

Let $(\mathrm{X}_1,\mathrm{Y}_1),\ldots,(\mathrm{X}_n,\mathrm{Y}_n)$ *be i.i.d. with* $p_{\mathrm{X},\mathrm{Y}}$ *and* $\gamma > 0$, $\epsilon > 0$,

(a) $\mathbf{P}\left((\mathrm{X}_1^n, \mathrm{Y}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathrm{X},\mathrm{Y})\right) \geq 1-\epsilon$ *for n large enough.* *W.L.L.N*

(b) $\left|\mathcal{A}_\gamma^{(n)}(\mathrm{X},\mathrm{Y})\right| \leq 2^{n(H(\mathrm{X},\mathrm{Y})+\gamma)}$, (*)

(c) $\left|\mathcal{A}_\gamma^{(n)}(\mathrm{X},\mathrm{Y})\right| \geq (1-\epsilon)2^{n(H(\mathrm{X},\mathrm{Y})-\gamma)}$ *for n large enough,* (*), *part* (a)

(d) *If* $(\mathrm{X}_1,\mathrm{Z}_1),\ldots,(\mathrm{X}_n,\mathrm{Z}_n)$ *be i.i.d. with* $p_{\mathrm{X},\mathrm{Z}}(x,z) = p_{\mathrm{X}}(x)\,p_{\mathrm{Y}}(z)$ *then*

$\mathbf{P}\left((\mathrm{X}_1^n, \mathrm{Z}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathrm{X},\mathrm{Y})\right) \leq 2^{-n(I(\mathrm{X};\mathrm{Y})-3\gamma)}$ *and* (*), *part* (b)

$\mathbf{P}\left((\mathrm{X}_1^n, \mathrm{Z}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathrm{X},\mathrm{Y})\right) \geq (1-\epsilon)2^{-n(I(\mathrm{X};\mathrm{Y})+3\gamma)}$. (*), *part* (c)

Proof Of Achievability I

- Encoding function $\varphi : \{1, \ldots, \lceil 2^{nR} \rceil\} \to \mathcal{X}_1^n$
- $\varphi_t : \{1, \ldots, \lceil 2^{nR} \rceil\} \to \mathcal{X}$ for all $t \in \{1, \ldots, n\}$ such that $\varphi(m) = [\varphi_1(m), \varphi_2(m), \ldots, \varphi_n(m)]$ for all $m \in \{1, \ldots, \lceil 2^{nR} \rceil\}$
- Let us consider a random choice of the encoding function s.t. $\mathbf{P}(\Phi = \varphi) = \prod_{m=1}^{\lceil 2^{nR} \rceil} \prod_{t=1}^{n} p_{\mathsf{X}}(\varphi_t(m))$ for some p_{X}.
- Decoding function $\widetilde{\varphi}$ is chosen based on φ as follows:

$$\widetilde{\varphi}(y_1^n) = \begin{cases} m & \text{if } \exists! m \in \{1, \ldots, \lceil 2^{nR} \rceil\} \text{ s.t. } (\varphi(m), y_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathsf{X}, \mathsf{Y}) \\ 1 & \text{otherwise} \end{cases}$$

- $\mathcal{E} := \{\omega : \mathsf{M} \neq \widetilde{\Phi}(\mathsf{Y}_1^n)\}$ then $\mathcal{E} \subset \mathcal{E}_1 \cup \mathcal{E}_2$ where

$$\mathcal{E}_1 := \{\omega : (\Phi(\mathsf{M}), \mathsf{Y}_1^n) \notin \mathcal{A}_\gamma^{(n)}(\mathsf{X}, \mathsf{Y})\}$$

$$\mathcal{E}_2 := \{\omega : \exists \widetilde{m} \neq \mathsf{M} \text{ s.t. } (\Phi(\widetilde{m}), \mathsf{Y}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathsf{X}, \mathsf{Y})\}$$

- $\mathbf{E}\left[P_{\mathsf{e}}^{(n)}\right] \leq \mathbf{P}(\mathcal{E}_1) + \mathbf{P}(\mathcal{E}_2)$ by the union bound.

Proof Of Achievability II

- Since $\mathcal{E}_1 := \{\omega : (\Phi(\mathsf{M}), \mathsf{Y}_1^n) \notin \mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y})\}$

$$\mathbf{P}(\mathcal{E}_1) = \sum_{m=1}^{\lceil 2^{nR} \rceil} \frac{1}{\lceil 2^{nR} \rceil} \underbrace{\mathbf{P}\Big((\Phi(m), \mathsf{Y}_1^n) \notin \mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y}) \Big| \mathsf{M} = m\Big)}_{\leq \epsilon \quad \text{for all } n \text{ large enough by AEP}}$$

$$\leq \epsilon \qquad \text{for all } n \text{ large enough.}$$

- Since $\mathcal{E}_2 := \{\omega : \exists \widetilde{m} \neq \mathsf{M} \text{ s.t. } (\Phi(\widetilde{m}), \mathsf{Y}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y})\}$

$$\mathbf{P}(\mathcal{E}_2) = \sum_{m=1}^{\lceil 2^{nR} \rceil} \frac{1}{\lceil 2^{nR} \rceil} \underbrace{\mathbf{P}\Big(\exists \widetilde{m} \neq m \text{ s.t. } (\Phi(\widetilde{m}), \mathsf{Y}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y}) \Big| \mathsf{M} = m\Big)}_{\leq \sum_{\widetilde{m} \neq m} \underbrace{\mathbf{P}\Big((\Phi(\widetilde{m}), \mathsf{Y}_1^n) \in \mathcal{A}_\gamma^{(n)}(\mathsf{X},\mathsf{Y}) \Big| \mathsf{M} = m\Big)}_{\leq 2^{-n(I(\mathsf{X};\mathsf{Y}) - 3\gamma)}}}$$

$$\leq \epsilon \qquad \text{for all } n \text{ large enough iff } R < I(\mathsf{X};\mathsf{Y}) \text{ and } \gamma < \frac{I(\mathsf{X};\mathsf{Y}) - R}{3}.$$

Proof Of Achievability III

- If $\mathbf{E}\left[P_{\mathbf{e}}^{(n)}\right] \leq 2\epsilon$ there exists at least one code such that $P_{\mathbf{e}}^{(n)} \leq 2\epsilon$.
- At least half of the messages in the code satisfying $P_{\mathbf{e}}^{(n)} \leq 2\epsilon$ satisfies $P_{\mathbf{e}}^{m} \leq 4\epsilon$ by Markov's inequality because $P_{\mathbf{e}}^{(n)} = \sum_{m=1}^{\lceil 2^{nR} \rceil} \frac{1}{\lceil 2^{nR} \rceil} P_{\mathbf{e}}^{m}$.
- For $\epsilon > 0$, $p_{\mathsf{X}} \in \mathcal{P}(\mathcal{X})$, and $R < I(\mathsf{X};\mathsf{Y})$ there exists a code $(\lceil 2^{nR-1} \rceil, n)$ code such that $P_{\mathbf{e}}^{(n),\max} \leq 4\epsilon$.
- In order establish the achievability of the rate $C\left(p_{\mathsf{Y}|\mathsf{X}}\right) - \delta$ we choose a p_{X} that satisfies $I(\mathsf{X};\mathsf{Y}) > C\left(p_{\mathsf{Y}|\mathsf{X}}\right) - \delta/2$.

Remark

For channels with finite input alphabets we have a p_{X} that satisfies $I(\mathsf{X};\mathsf{Y}) = C\left(p_{\mathsf{Y}|\mathsf{X}}\right)$. This observation is, however, not necessary to establish the achievability of any rate R satisfying $R < C\left(p_{\mathsf{Y}|\mathsf{X}}\right)$.

J. H. B. Kemperman.
On the Shannon capacity of an arbitrary channel.
Indagationes Mathematicae (Proceedings), 77(2):101–115, 1974.